

PHYS 380 - HW 2

P.6-8

a) Displacement current: $\vec{J}_D = \frac{\partial \vec{D}}{\partial t}$ assuming homogenous
isotropic linear medium $\vec{J}_D = \epsilon_r \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

Conduction current: $\vec{J}_C = \sigma \vec{E}$

In terms of phasor $\begin{cases} \vec{J}_{D-s} = \epsilon_r \epsilon_0 j\omega \vec{E}_s \\ \vec{J}_{C-s} = \sigma \vec{E}_s \end{cases}$

$$\frac{|\vec{J}_{D-s}|}{|\vec{J}_{C-s}|} = \frac{\epsilon_r \epsilon_0 \omega}{\sigma} \approx \frac{2\pi \times 10^{11} \times \frac{1}{36\pi} \times 10^{-9}}{5.70 \times 10^7} \approx \boxed{9.75 \times 10^{-8}}$$

b) $\nabla \times \vec{H} = \vec{J} = \sigma \vec{E} \Rightarrow$ neglecting the displacement current in a good conductor

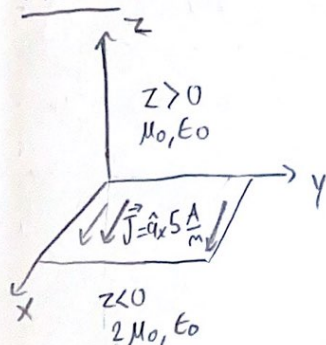
$$\nabla \times (\nabla \times \vec{H}) = \sigma (\nabla \times \vec{E}) = \sigma \left(-\frac{\partial \vec{B}}{\partial t} \right) = -\sigma \mu \frac{\partial \vec{H}}{\partial t}$$

Using $\nabla \times (\nabla \times \vec{H}) = \nabla(\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = -\nabla^2 \vec{H}$
 0 since $\nabla \cdot \vec{B} = 0$

Then it follows: $\boxed{\nabla^2 \vec{H} - \sigma \mu \frac{\partial \vec{H}}{\partial t} = 0}$

OR in terms of phasor quantities: $\boxed{\nabla^2 \vec{H}_s - j\omega \sigma \mu \vec{H}_s = 0}$

P.6-9



$$\vec{H}_1 = a_x 30 + a_y 40 + a_z 20 \frac{A}{m}$$

Boundary Conditions for Electromagnetism

(1) $\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J} \Leftrightarrow H_{1t} - H_{2t} = J_s$

(2) $B_{1n} = B_{2n} \Leftrightarrow \mu_1 H_{1n} = \mu_2 H_{2n}$ Transverse direction is orthogonal to current density

(3) $E_{1t} = E_{2t}$

(4) $D_{1n} - D_{2n} = \rho_s$

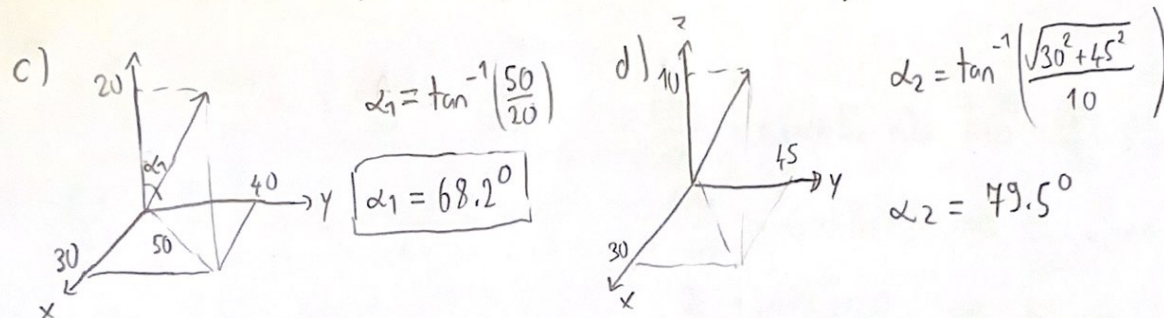
* Using these eqns and $\vec{B} = \mu \vec{H}$, we find

$$|\hat{a}_x \times (H_{2x} - (\hat{a}_x 30 + \hat{a}_y 40))| = 5 \Rightarrow H_{2y} - H_{1y} = 5 \Rightarrow H_{2y} = 45$$

$$H_{2x} = H_{1x} \text{ (no condition)} \quad \text{and} \quad \mu_1 H_{1z} = \mu_2 H_{2z} \Rightarrow H_{2z} = \frac{\mu_0}{2\mu_0} 20 = 10$$

$$a) \vec{H}_2 = \hat{a}_x 30 + \hat{a}_y 45 + \hat{a}_z 10 \text{ A/m}$$

$$b) \vec{B}_2 = \mu_2 \vec{H}_2 = 2\mu_0 \vec{H}_2 = (\hat{a}_x 60 + \hat{a}_y 90 + \hat{a}_z 20) \times 4\pi \times 10^{-7} \text{ T}$$



P.6-10

* Inside a magnetic material with infinite permeability, we should have $\vec{H}_{\text{mag}} = 0$ since $\lim_{\mu \rightarrow \infty} \vec{B} = \lim_{\mu \rightarrow \infty} \mu \vec{H}_{\text{mag}}$ must be finite.

Let medium 1 be free space (ϵ_0, μ_0) and medium 2 be magnetic material $\Rightarrow \vec{H}_2 = 0$

Boundary Conditions

$$\begin{cases} (1) H_{1t} = J_s \text{ or } \hat{a}_{n1} \times \vec{H}_1 = \vec{J}_s & (2) B_{1n} = B_{2n} \\ (3) E_{1t} = E_{2t} & (4) D_{1n} - D_{2n} = \rho_s \text{ or } \hat{a}_{n2} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \end{cases}$$

P.6-11

Let $u(R, t) = f(t - R\sqrt{\mu\epsilon}) = f(\phi)$ where $\phi = t - R\sqrt{\mu\epsilon}$

Homogeneous wave eq: $\frac{\partial^2 u}{\partial R^2} - \mu\epsilon \frac{\partial^2 u}{\partial t^2} = 0$

$$\frac{\partial^2 u}{\partial R^2} = \frac{\partial}{\partial R} \left(-\frac{\partial \phi}{\partial R} f'(\phi) \right) = -\sqrt{\mu\epsilon} \frac{\partial f'(\phi)}{\partial R} = -\sqrt{\mu\epsilon} (-\sqrt{\mu\epsilon}) f''(\phi) = \mu\epsilon f''(\phi)$$

$$\frac{\partial^2 u}{\partial t^2} = f''(\phi) \text{ since } \frac{\partial \phi}{\partial t} = 1$$

Then we see that $\mu\epsilon f''(\phi) - \mu\epsilon f''(\phi) = 0 \Rightarrow$ wave eq. satisfied.

P.6-12

Time harmonic Maxwell
Equations in terms of phasors

$$\left\{ \begin{array}{l} \textcircled{1} \nabla \times \vec{E} = -j\omega\mu\vec{H} \\ \textcircled{2} \nabla \times \vec{H} = \vec{J} + j\omega\epsilon\vec{E} \end{array} \right. , \quad \left\{ \begin{array}{l} \textcircled{3} \nabla \cdot \vec{E} = \rho_v/\epsilon \\ \textcircled{4} \nabla \cdot \vec{H} = 0 \end{array} \right.$$

a) $\vec{E} = \vec{E}(z)$, $\vec{H} = H(z)$,

$\textcircled{1} \nabla \times \vec{E} = -\frac{\partial E_y}{\partial z} \hat{a}_x + \frac{\partial E_x}{\partial z} \hat{a}_y \Rightarrow$

$$\frac{\partial E_y}{\partial z} = j\omega\mu H_x , \quad \frac{\partial E_x}{\partial z} = -j\omega\mu H_y$$

$$H_z = 0$$

$\textcircled{2} \nabla \times \vec{H} = -\frac{\partial H_y}{\partial z} \hat{a}_x + \frac{\partial H_x}{\partial z} \hat{a}_y \Rightarrow$

$$-\frac{\partial H_y}{\partial z} = +j\omega\epsilon E_x + J_x \xrightarrow{J=0} -\frac{\partial H_y}{\partial z} = j\omega\epsilon E_x$$

$\textcircled{3} \frac{\partial E_z}{\partial z} = \frac{\rho_e}{\epsilon} \Rightarrow E_z = \frac{\rho_e z}{\epsilon} + \text{const}$

$$\frac{\partial H_x}{\partial z} = +j\omega\epsilon E_y + J_y \Rightarrow \frac{\partial H_x}{\partial z} = j\omega\epsilon E_y$$

$\textcircled{4} \frac{\partial H_z}{\partial z} = 0 \Rightarrow H_z = \text{const}$

$$J_z + j\omega\epsilon E_z = 0 \Rightarrow E_z = 0$$

b) $\nabla \times \vec{V} = \frac{1}{R\sin\theta} \left[\frac{\partial}{\partial\theta} (\sin\theta V_\phi) - \frac{\partial V_\theta}{\partial\phi} \right] \hat{a}_R + \frac{1}{R} \left[\frac{1}{\sin\theta} \frac{\partial V_R}{\partial\phi} - \frac{\partial}{\partial R} (R V_\theta) \right] \hat{a}_\theta$
 $+ \frac{1}{R} \left[\frac{\partial}{\partial R} (R V_\theta) - \frac{\partial V_R}{\partial\theta} \right] \hat{a}_\phi = -\frac{1}{R} \frac{\partial (R V_\phi)}{\partial R} \hat{a}_\theta + \frac{1}{R} \frac{\partial (R V_\theta)}{\partial R} \hat{a}_\phi ,$

$$\nabla \cdot \vec{V} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 V_R) + \frac{1}{R\sin\theta} \frac{\partial}{\partial\theta} (\sin\theta V_\theta) + \frac{1}{R\sin\theta} \frac{\partial V_\phi}{\partial\phi}$$

$$= \frac{2}{R} V_R + \frac{\partial V_R}{\partial R} + \frac{\cos\theta V_\theta}{R\sin\theta}$$

$\textcircled{1} +\frac{1}{R} E_\phi + \frac{\partial E_\phi}{\partial R} = +j\omega\mu H_\theta , \quad \frac{1}{R} E_\theta + \frac{\partial E_\theta}{\partial R} = -j\omega\mu H_\phi , \quad H_R = 0$

$\textcircled{2} \text{ Assuming } J=0 \quad \frac{1}{R} H_\phi + \frac{\partial H_\phi}{\partial R} = -j\omega\mu E_\theta , \quad \frac{1}{R} H_\theta + \frac{\partial H_\theta}{\partial R} = j\omega\mu E_\phi , \quad E_R = 0$

$\textcircled{3} \frac{2E_R}{R} + \frac{\partial E_R}{\partial R} + \frac{\cos\theta E_\theta}{R\sin\theta} = \frac{\rho_v}{\epsilon}$

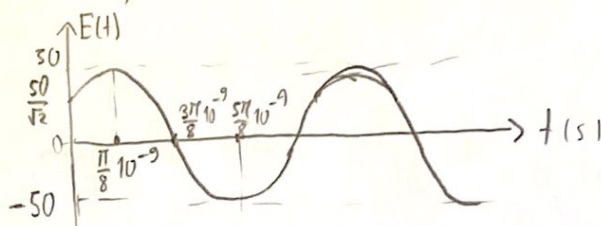
$\textcircled{4} \frac{2H_R}{R} + \frac{\partial H_R}{\partial R} + \frac{\cos\theta H_\theta}{R\sin\theta} = 0$

P-6.13

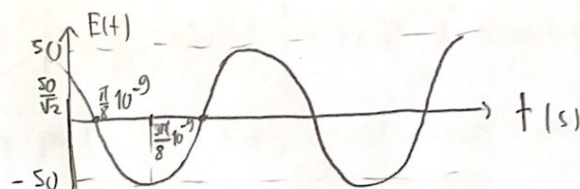
$$E(z, t) = 50 \cos(2\pi 10^9 t - kz) \frac{V}{m}$$

$$k = \frac{2\pi}{\lambda}, f = 10^9, \omega = 2\pi 10^9$$

$$\begin{aligned} a) E(t) &= 50 \cos\left(2\pi 10^9 t - \frac{100.125\lambda}{\lambda} \times 2\pi\right) \\ &= 50 \cos\left(2\pi 10^9 t - \frac{\pi}{4}\right) \end{aligned}$$

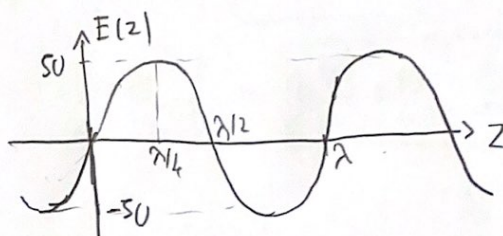


$$b) E(t) = 50 \cos\left(2\pi 10^9 t + \frac{\pi}{4}\right)$$



$$c) f = 10^9, T = 10^{-9} \Rightarrow \text{at } t = \frac{T}{4}$$

$$\begin{aligned} E(z) &= 50 \cos\left(\frac{\pi}{2} - kz\right) \\ &= 50 \sin(kz) \\ &= 50 \sin\left(\frac{2\pi}{\lambda} z\right) \end{aligned}$$



P-6.14

$$\vec{E}(z, t) = a_x E_0 \cos\left[10^8 \pi \left(t - \frac{z}{c}\right) + \psi\right]$$

$$\vec{E}_1(z, t) = a_x 0.03 \sin\left[10^8 \pi \left(t - \frac{z}{c}\right)\right], \vec{E}_2(z, t) = a_x 0.04 \cos\left[10^8 \pi \left(t - \frac{z}{c}\right) - \frac{\pi}{3}\right]$$

Let's define the phasor of $\vec{E}(z, t)$ as $\vec{E}(z, t) = a_x \text{Re}\{E_0 e^{j\psi} e^{j10^8 \pi (t - \frac{z}{c})}\}$

$$\begin{aligned} \text{Then } E_s &= E_0 e^{j\psi}, E_{1s} = 0.03 e^{-j\frac{\pi}{2}}, E_{2s} = 0.04 e^{-j\frac{\pi}{3}} \\ &\Rightarrow \text{scalar Phasor } E_s \end{aligned}$$

We should have $E_s = E_{1s} + E_{2s}$ since all the vector quantities are in a_x direction

$$E_{1s} = 0.03 \cos\left(\frac{\pi}{2}\right) - j 0.03 \sin\left(\frac{\pi}{2}\right) = -j 0.03, E_{2s} = 0.04 \cos\left(\frac{\pi}{3}\right) - j 0.04 \sin\left(\frac{\pi}{3}\right) = 0.02 - j 0.02\sqrt{3}$$

$$\text{Then } E_s = E_{1s} + E_{2s} = 0.02 - j(0.03 + 0.02\sqrt{3}) = 0.02 - j 0.064641 = E_0 e^{j\psi}$$

$$E_0 = \sqrt{0.02^2 + (0.064641)^2} = 0.068 \frac{V}{m}, \psi = \tan^{-1}\left(\frac{-0.064641}{0.02}\right) = -72.8^\circ$$

P.6-15

$$\vec{H}(R,t) = a_\phi H_0 \cos(\omega t - kR) \xrightarrow{\text{phasor}} \vec{H}_s = H_0 a_\phi$$

$$\vec{H}_1(R,t) = a_\phi 10^{-4} \sin(\omega t - kR) \longrightarrow \vec{H}_{s1} = 10^{-4} e^{-j\frac{\pi}{2}} a_\phi$$

$$\vec{H}_2(R,t) = a_\phi 2 \times 10^{-4} \cos(\omega t - kR + \alpha) \longrightarrow \vec{H}_{s2} = 2 \times 10^{-4} e^{j\alpha} a_\phi$$

$$\vec{H}_s = \vec{H}_{s1} + \vec{H}_{s2} \implies H_0 = 10^{-4} \left[\cos\left(\frac{\pi}{2}\right) - j \sin\left(\frac{\pi}{2}\right) \right] + 2 \times 10^{-4} [\cos(\alpha) + j \sin(\alpha)]$$

→ since H_0 is purely real, imaginary parts on RHS must cancel out. Then,

$$-10^{-4} = 2 \times 10^{-4} \sin(\alpha) \implies \sin(\alpha) = -\frac{1}{2} \implies \boxed{\alpha = -30^\circ}$$

$$H_0 = 10^{-4} \underbrace{\cos\left(\frac{\pi}{2}\right)}_0 + 2 \times 10^{-4} \cos(-30^\circ) = \sqrt{3} \times 10^{-4} \frac{A}{m} = \boxed{1.73 \times 10^{-4} \frac{A}{m}}$$

P.6-16

$$a) \quad \nabla \times \vec{E} = -j\omega \mu \vec{H}, \quad \nabla \times (\nabla \times \vec{E}) = -j\omega \mu (\nabla \times \vec{H}) = -j\omega \mu \left(\vec{J} + \underbrace{j\omega \vec{D}}_{\frac{\partial \vec{D}}{\partial t} \text{ in harmonic media}} \right)$$

$$\text{But also } \nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = \nabla\left(\frac{\rho_v}{\epsilon}\right) - \nabla^2 \vec{E}$$

$$\nabla\left(\frac{\rho_v}{\epsilon}\right) - \nabla^2 \vec{E} = -j\omega \mu \vec{J} + \omega^2 \mu \epsilon \vec{E} \quad (\text{assuming homogeneous, linear, isotropic media})$$

$$\nabla^2 \vec{E} + (\omega \sqrt{\mu \epsilon})^2 \vec{E} = j\omega \mu \vec{J} + \nabla\left(\frac{\rho_v}{\epsilon}\right) \implies \boxed{\nabla^2 \vec{E} + k^2 \vec{E} = j\omega \mu \vec{J} + \nabla\left(\frac{\rho_v}{\epsilon}\right)}$$

$$\rightarrow \omega = 2\pi f, \quad \sqrt{\mu \epsilon} = \frac{1}{u_p} \implies \omega \sqrt{\mu \epsilon} = 2\pi \frac{f}{u_p} = \frac{2\pi}{\lambda} = k \quad \text{most general form}$$

$$b) \quad \nabla \times \vec{H} = \vec{J} + j\omega \epsilon \vec{E}, \quad \nabla \times (\nabla \times \vec{H}) = \nabla \times \vec{J} + j\omega \epsilon \nabla \times \vec{E}$$

$$\nabla \times \vec{E} = -j\omega \mu \vec{H} \quad \text{and} \quad \nabla \times (\nabla \times \vec{H}) = \nabla\left(\underbrace{\nabla \cdot \vec{H}}_0\right) - \nabla^2 \vec{H} = -\nabla^2 \vec{H}$$

$$-\nabla^2 \vec{H} = \nabla \times \vec{J} + j\omega \epsilon (-j\omega \mu \vec{H}) \implies \nabla^2 \vec{H} + \omega^2 \mu \epsilon \vec{H} = -\nabla \times \vec{J}$$

$$\omega^2 \mu \epsilon = k^2 \implies \boxed{\nabla^2 \vec{H} + k^2 \vec{H} = -\nabla \times \vec{J}}$$

$\updownarrow$
 shown above wave number

P.6-17

a) $\vec{A}(R) = \frac{\mu}{4\pi} \int_{V'} \frac{\vec{J} e^{-jkR}}{R} dV \Rightarrow$ phasor retarded vector potential

$\rightarrow$ Note that current distribution consist of just an infinitesimal cable of length dl :

$$\vec{J} dV = I_0 \cos \omega t dl$$

$$\vec{A}(R) = \frac{\mu dl \cdot I_0 \cos \omega t}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \begin{cases} A_R = A_z \cos \theta = \frac{\mu_0 I_0 dl \cos \omega t}{4\pi} \left(\frac{e^{-jkR}}{R} \cos \theta \right) \\ A_\theta = A_z \sin \theta = \frac{\mu_0 I_0 dl \cos \omega t}{4\pi} \left(\frac{e^{-jkR}}{R} \sin \theta \right) \\ A_\phi = 0 \end{cases}$$

b) $\mu \vec{H} = \nabla \times \vec{A} = \hat{a}_\phi \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \Rightarrow$ other terms in curl cancel either due to $A_\phi = 0$ or because A_R, A_θ are not a function of ϕ .

$$\vec{H} = \frac{1}{\mu} \hat{a}_\phi \frac{1}{R} \left[\frac{\mu dl I_0 \cos \omega t (-jk)}{4\pi} e^{-jkR} \sin \theta - \frac{\mu_0 dl I_0 \cos \omega t}{4\pi} \frac{e^{-jkR}}{R} (-\sin \theta) \right]$$

$$\boxed{\vec{H} = -\hat{a}_\phi \frac{I_0 dl \cos \omega t}{4\pi} \left(\frac{e^{-jkR}}{R} \sin \theta \right) \left[jk + \frac{1}{R} \right]}$$

P.16-18 $\vec{E} = \hat{a}_R \frac{E_0}{R} e^{-jkz}$ (V/m) between conducting layers

a) $k = \frac{2\pi}{\lambda} = \frac{2\pi f}{c} = \frac{2\pi \times 60 \times 10^6}{3 \times 10^8} = 0.4\pi \text{ rad/m}$

b) $\nabla \times \vec{E} = -j\omega\mu \vec{H} = \frac{\partial E_R}{\partial z} \hat{a}_\phi \Rightarrow$ terms depending on ϕ and z are eliminated in the curl of cylindrical coordinates

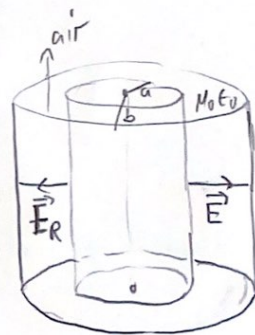
$$= +\hat{a}_\phi \frac{E_0}{R} (-jk) e^{-jkz} \Rightarrow \vec{H} = \hat{a}_\phi \frac{E_0 k}{\omega \mu_0 R} e^{-jkz}$$

$$\boxed{\vec{H} = \hat{a}_\phi \frac{E_0}{120\pi R} e^{-jkz}}$$

c) Surface currents induced in the conductor is

Inner conductor $\Rightarrow \vec{J}_s(R=a) = \hat{a}_z \frac{E_0}{120\pi a} e^{-j0.4\pi z}$

Outer conductor $\Rightarrow \vec{J}_s(R=b) = -\hat{a}_z \frac{E_0}{120\pi b} e^{-j0.4\pi z}$



surface normal vector

$$\vec{J}_s = \hat{n} \times \vec{H} = \underbrace{\hat{a}_R \times \hat{a}_\phi}_{\pm \hat{a}_z} \frac{E_0}{120\pi R} e^{-jkz}$$

$\rightarrow$ Inner surf: $\hat{n} = \hat{a}_R$

$\rightarrow$ Outer surf: $\hat{n} = -\hat{a}_R$

$$\frac{\omega}{k} = c$$

$$\mu_0 c = 4\pi \times 10^{-7} \times 3 \times 10^8$$

P.6-19

$$E(R, \theta; t) = \hat{a}_\theta \frac{10^{-3}}{R} \sin \theta \cos(2\pi 10^9 t - kR)$$

→ In free space $\mu = \mu_0$, $\epsilon = \epsilon_0 \Rightarrow u_p = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{\omega}{k}$

$$k = \frac{2\pi \times 10^9 \text{ rad/s}}{3 \times 10^8 \text{ m/s}} = \boxed{20.94 \text{ rad/m}}$$

$$\nabla \times \vec{E} = -j\omega \mu \vec{H}, \quad \nabla \times \vec{E} = -\hat{a}_\phi \frac{1}{R} \frac{\partial (RA_\theta)}{\partial R}, \quad \left(\begin{array}{l} \text{eliminating } E_R, E_\phi \text{ terms and all} \\ \text{the terms that does depend on } \phi \end{array} \right)$$

In phasor form: $\vec{E}_s = \hat{a}_\theta \frac{10^{-3}}{R} \sin \theta \angle -kR$

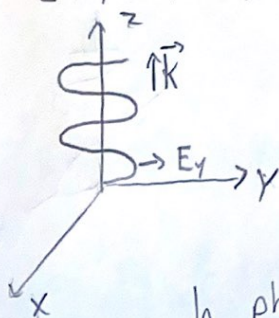
$$\nabla \times \vec{E}_s = \hat{a}_\phi \frac{1}{R} \frac{\partial}{\partial R} \left(10^{-3} \sin \theta e^{jkR} \right) = \hat{a}_\phi \frac{1}{R} 10^{-3} \sin \theta (-jk) e^{-jkR}$$

$$\vec{H}_s = \frac{\nabla \times \vec{E}_s}{-j\omega \mu_0} = \hat{a}_\phi \frac{10^{-3}}{\frac{\omega}{k} \mu_0 R} \sin \theta e^{-jkR} = \hat{a}_\phi \frac{10^{-3}}{120\pi R} \sin \theta e^{-jkR}$$

$$\boxed{\vec{H}(R, \theta; t) = \hat{a}_\phi \frac{10^{-3}}{120\pi R} \sin \theta \cos(2\pi 10^9 t - \frac{20\pi R}{3}) \frac{A}{m}}$$

P.6-20

$$\vec{E}(x, z, t) = \hat{a}_y \times 0.1 \sin(10\pi x) \cos(6\pi 10^9 t - \beta z)$$



* Wave is travelling in $\hat{a}_z$ direction with its electric field oscillating in $\hat{a}_y$ direction.

→ Wave number is not simply $\frac{\omega}{c}$ in this case due to $\sin(10\pi x)$ term, suggesting an inhomogeneous media or waveguide.

In phasor form $\vec{E}_s = \hat{a}_y 0.1 \sin(10\pi x) e^{-j\beta z}$

$$\begin{aligned} \nabla \times \vec{E}_s &= \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_{sy} & 0 \end{vmatrix} = -\hat{a}_x \frac{\partial}{\partial z} \left(0.1 \sin(10\pi x) e^{-j\beta z} \right) + \hat{a}_z \frac{\partial}{\partial x} \left(0.1 \sin(10\pi x) e^{-j\beta z} \right) \\ &= \hat{a}_x j\beta 0.1 \sin(10\pi x) e^{-j\beta z} + \hat{a}_z \pi \cos(10\pi x) e^{-j\beta z} \end{aligned}$$

$$\nabla \times \vec{E}_s = -j\omega\mu_0 \vec{H}_s$$

$$\vec{H}_s = -\hat{a}_x \frac{0.1\beta}{\mu_0\omega} \sin(10\pi x) e^{-j\beta z} + \hat{a}_z \frac{\pi}{\mu_0\omega} \cos(10\pi x) (j) e^{-j\beta z}$$

$$\nabla \times \vec{H}_s = +\hat{a}_y \frac{j 0.1\beta^2}{\mu_0\omega} \sin(10\pi x) e^{-j\beta z} + \hat{a}_y \frac{j 10\pi^2}{\mu_0\omega} \sin(10\pi x) e^{-j\beta z}$$

$$\vec{E}_s = \frac{\nabla \times \vec{H}_s}{j\omega\epsilon_0} = \hat{a}_y \frac{0.1}{\omega^2\mu_0\epsilon_0} [\beta^2 + (10\pi)^2] \sin(10\pi x) e^{-j\beta z}$$

But $\vec{E}_s$ also equals to $\vec{E}_s = \hat{a}_y 0.1 \sin(10\pi x) e^{-j\beta z}$

$$\beta^2 = \omega^2\mu_0\epsilon_0 - (10\pi)^2 = \left(\frac{6\pi \times 10^9}{3 \times 10^8}\right)^2 - 100\pi^2 = 300\pi^2 \Rightarrow \boxed{\beta = \sqrt{300}\pi = 54.4 \text{ rad/m}}$$

$$\vec{H}_s = -\hat{a}_x \frac{0.1 \times 54.4}{4\pi \times 10^{-7} \times 6\pi \times 10^9} \sin(10\pi x) e^{-j54.4 z} + \hat{a}_z \frac{\pi \cos(10\pi x)}{4\pi \times 10^{-7} \times 6\pi \times 10^9} e^{\frac{\pi}{2} - j\beta z}$$

$$\vec{H}_s(x, z, t) = -\hat{a}_x 2.2967 \times 10^{-4} \sin(10\pi x) \cos(6\pi \times 10^9 t - 54.4 z)$$

$$+ \hat{a}_z 1.3263 \times 10^{-4} \cos(10\pi x) \sin(6\pi \times 10^9 t - 54.4 z) \frac{A}{m}$$

phasor form
↓
 $e^{-j\beta z - \frac{\pi}{2}}$

P.6:21 $\vec{H}(x, z, t) = \hat{a}_y 2 \cos(15\pi x) \sin(6\pi \times 10^9 t - \beta z) A/m$ or $\vec{H}_s = \hat{a}_y 2 \cos(15\pi x) e^{-j\beta z - \frac{\pi}{2}}$

$$\vec{E}_s = \frac{1}{j\omega\epsilon} \nabla \times \vec{H}_s = \frac{2}{j\omega\epsilon} [\hat{a}_x j\beta \cos(15\pi x) - \hat{a}_z 15\pi \sin(15\pi x)] e^{-j\beta z - \frac{\pi}{2}}$$

$$\vec{H}_s = -\frac{1}{j\omega\mu} = \frac{2}{\omega^2\mu\epsilon} \hat{a}_y [\beta^2 \cos(15\pi x) + (15\pi)^2 \cos(15\pi x)] e^{-j\beta z - \frac{\pi}{2}} = 2\hat{a}_y \cos(15\pi x) e^{-j\beta z - \frac{\pi}{2}}$$

$$\beta^2 + (15\pi)^2 = \omega^2\mu\epsilon = \left(\frac{6\pi \times 10^9}{3 \times 10^8}\right)^2 = 400\pi^2 \Rightarrow \beta = \sqrt{45}\pi = \boxed{41.6 \text{ rad/m}}$$

$$\vec{E}_s = \left[\hat{a}_x \frac{2\beta}{\omega\epsilon_0} \cos(15\pi x) + \hat{a}_z \frac{30\pi}{\omega\epsilon_0} \sin(15\pi x) j \right] e^{-j\beta z - \frac{\pi}{2}}$$

$$= \hat{a}_x \frac{83.2}{6\pi \times 10^9 \times 8.85 \times 10^{-12}} \cos(15\pi x) e^{-j\beta z - \frac{\pi}{2}} + \hat{a}_z \frac{30\pi}{6\pi \times 8.85 \times 10^{-12}} \sin(15\pi x) e^{-j\beta z}$$

$$\vec{E}(x, z, t) = \hat{a}_x 498.7 \cos(15\pi x) \sin(6\pi \times 10^9 t - 41.6 z)$$

$$+ \hat{a}_z 564.97 \sin(15\pi x) \cos(6\pi \times 10^9 t - 41.6 z) V/m$$